EE4-40
EE9-CS7-26
EE9-SO20

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2021

MSc and EEE/EIE PART IV: MEng and ACGI

INFORMATION THEORY

Tuesday, 11 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : D. Gunduz

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Information for students

Notation:

(a) Random variables are shown in Tahoma font. x, $\mathbf{x}$, $\mathbf{X}$ denote a random scalar, vector and matrix respectively.

(b) The size of a set A is denoted by $|A|$.

(c) $\oplus$ denotes the exclusive-or operation, or modulo-2 addition.

(d) "i.i.d." means "independent identically distributed".

The Questions

1. Basics.

a) Suppose x_1 and x_2 are i.i.d. Bernoulli random variables, taking values of 0 and 1 with probabilities $p = 0.2$ and $q = 1 - p = 0.8$, respectively. Let $y = \min(x_1, x_2)$. Compute the following entropy or mutual information values:

i) $H(y)$ [3]

ii) $I(x_1 ; y)$ [4]

iii) $I(x_{1:2} ; y)$ [3]

b) The random process $\{x_n\}$ is a sequence of i.i.d. standard normal random variables. Let

$$y_n = \sum_{i=0}^{\infty} 2^{-i} x_{n-i}$$

i) Find the mutual information $I(x_n; y_n)$. [10]

ii) Find the differential entropy rate of the random process $\{y_n\}$. [5]

Note: The differential entropy rate is defined similarly to the entropy rate of a discrete random process, except that entropy is replaced with differential entropy. That is,

$$h(\boldsymbol{x}) = \lim_{n\to\infty} \frac{1}{n} h(x_{1:n}) = \lim_{n\to\infty} h(x_n \mid x_{1:n-1})$$

if the limits exist.

2. Source and channel coding.

a) Lempel-Ziv coding. Consider the following sequence, by concatenating all binary strings of length 1, 2, 3, etc.

$$x = 0100011011000001010011\ldots$$

i) Give the LZ78 parsing and encoding for the sequence up to the last bit shown above. You may simply use numbers 1, 2, 3, … to represent the locations. [5]

ii) Calculate the number of encoding bits per symbol for this sequence as of length $n \to \infty$. [5]

b) Fano's inequality. Consider the Markov chain shown in Fig. 2.1, where x is the channel input, y is the channel output, and $\hat{x}$ is the estimate of x.

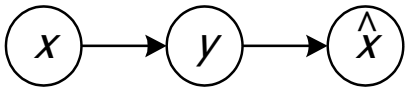


Fig. 2.1. Markov chain.

The input alphabet $X = \{0, 1, 2\}$, with probability mass vector $\mathbf{p}_x = [0.3, 0.5, 0.2]$. The channel has the following transitional matrix

$$\mathbf{Q} = \begin{bmatrix} 1/2 & 1/4 & 1/4 \\ 1/4 & 1/2 & 1/4 \\ 1/4 & 1/4 & 1/2 \end{bmatrix} \quad x, y \in \{0,1,2\}$$

i) Compute the error probability of the maximum a posteriori estimator. [5]

ii) Compute the conditional entropy $H(x \mid y)$. [5]

iii) Using Fano's inequality $H(x \mid y) \le 1 + p_e \log(|X| - 1)$, where P_e is the error probability, compute the Fano bound on the error probability, and compare with i). [5]

3. Network information theory.

a) Consider a two-user frequency-division multiple access (FDMA) channel with Gaussian noise. Assuming the signal powers P_1, P_2, noise power spectral density N and total bandwidth $W = W_1 + W_2$ are given. What is the bandwidth W_1 that maximizes the total capacity? [10]

b) Slepian-Wolf coding. Let Z_1 and Z_2 be Bernoulli(p) and suppose Z_1 and Z_2 are independent, where $P(Z_i = 0) = p$ and $P(Z_i = 1) = q$. Let $X = Z_1 Z_2$, and $Y = Z_1 - Z_2$.

Find the Slepian-Wolf rate region of achievable (R_X, R_Y). [15]

4. Typicality.

Consider discrete-valued random vectors $\mathbf{x}$ and $\mathbf{y}$ of length n, where each pair (x_i, y_i) is drawn i.i.d. from the joint probability distribution function $p_{xy}(x,y)$. Let $p_x(x)$ and $p_y(y)$ be the probability distribution functions of x_i and y_i, respectively. The jointly typical set is the set of vector pairs satisfying the following conditions:

$$J_\varepsilon^{(n)} = \Big\{\mathbf{x},\mathbf{y} : \Big|-n^{-1}\log_2 p_x(\mathbf{x}) - H(X)\Big| < \varepsilon, \\ \Big|-n^{-1}\log_2 p_y(\mathbf{y}) - H(Y)\Big| < \varepsilon, \\ \Big|-n^{-1}\log_2 p_{xy}(\mathbf{x},\mathbf{y}) - H(X,Y)\Big| < \varepsilon\Big\}$$

where, since the sequences are i.i.d., the probability $p_x(\mathbf{x}) = \prod_{i=1}^{n} p_x(x_i)$ and $p_y(\mathbf{y})$ and $p_{xy}(\mathbf{x},\mathbf{y})$ can be written in a similar fashion. Now consider joint typicality arising from the binary symmetric channel $y = x \oplus z$ shown in Fig. 4.1, where the cross-over probability is $p = P(z = 1) = 0.1$. Let the input distribution be the uniform distribution, i.e., $p_x(x) = (0.5, 0.5)$.

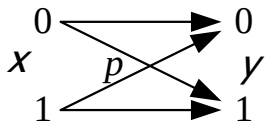


Fig. 4.1. Binary symmetric channel.

a) Calculate $H(x)$, $H(y)$, $H(x, y)$ and $I(x; y)$.

[5]

b) Of the 2^n possible input sequences of length n, how many of them are typical, i.e., member of $T_\varepsilon^{(n)}(\mathbf{x}) = \{x : \Big|-n^{-1}\log_2 p_x(\mathbf{x}) - H(X)\Big| < \varepsilon\}$ for ε = 0.1? How many typical sequences are in $T_\varepsilon^{(n)}(\mathbf{y})$?

[5]

c) Derive the condition that **z** is typical, and show that (**x**, **y**) being jointly typical is equivalent to the condition that **x** is typical and **z** is typical.

[5]

d) When n = 25, one obtains the table in the following page. Determine the size and probability of the jointly typical set $J_\varepsilon^{(n)}$ for $\varepsilon = 0.1$.

[5]

k	$\binom{n}{k}$	$\sum_{j\le k}\binom{n}{j}$	$p(x^n)=p^k(1-p)^{n-k}$	$\binom{n}{k}p^k(1-p)^{n-k}$	Cumul. pr.	$-\frac{1}{n}\log p(x^n)$
0	1	1	7.178975e-02	0.071790	0.071790	0.152003
1	25	26	7.976639e-03	0.199416	0.271206	0.278800
2	300	326	8.862934e-04	0.265888	0.537094	0.405597
3	2300	2626	9.847704e-05	0.226497	0.763591	0.532394
4	12650	15276	1.094189e-05	0.138415	0.902006	0.659191
5	53130	68406	1.215766e-06	0.064594	0.966600	0.785988
6	177100	245506	1.350851e-07	0.023924	0.990523	0.912785
7	480700	726206	1.500946e-08	0.007215	0.997738	1.039582
8	1081575	1807781	1.667718e-09	0.001804	0.999542	1.166379
9	2042975	3850756	1.853020e-10	0.000379	0.999920	1.293176
10	3268760	7119516	2.058911e-11	0.000067	0.999988	1.419973
11	4457400	11576916	2.287679e-12	0.000010	0.999998	1.546770
12	5200300	16777216	2.541865e-13	0.000001	0.999999	1.673567

e) Assume that 2^{nR} codewords are chosen uniformly over the 2^n possible binary sequences of length n. The decoder uses jointly typical decoding, i.e., it looks at the received sequence and tries to find a codeword in the code that is jointly typical with the received sequence. What is the decoding error probability?

[5]